

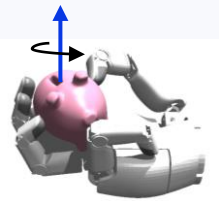
WM-Craftnet: World Synesthesia Model for Robust In-hand Manipulation

World Synesthesia Model Pretraining

In-hand Interaction in Simulation

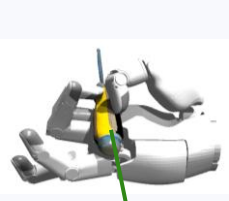
Rotation

piggy bank around z axis



Translation

screwdriver along y axis



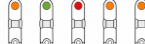
other skills
...

Inputs

Proprioception q_t



Tactile s_t



Wrist depth I_t



Prev. action u_{t-1}



Reward. r_t



World Synesthesia Model
(multi-modal RSSM)

deterministic state

$h_{t-1} \rightarrow h_t \rightarrow h_{t+1}$

$z_t \rightarrow z_t \rightarrow z_t$

Stochastic state

predictive feature h_t

...

Prediction Objectives

Reconstruct depth



Reconstruct proprio-tactile



Predict reward



Reason object pose, shape, contact

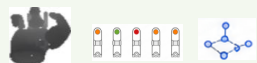


Reusable Synesthetic Prior

World Synesthesia Model
(pretrained)



Reconstruct per modality



Predict object motion



Reason about dynamics, geometry and contact



Synesthetic prior



Reusable knowledge



Knowledge transfer to new tasks

Depth Denoising



Scalable Object Types



WSM-conditioned Policy

Current obs.



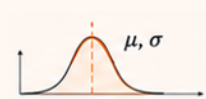
Commanded axis



WSM features



Asymmetric Actor-Critic Policy

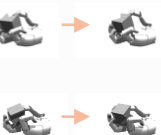


Relative action u_t

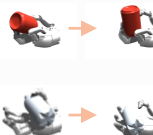
Update Target
 $q_{target,t} = q_{target,t-1} + \alpha u_t$

Outcomes

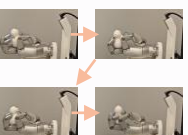
Unseen Objects



Perturbation Recovery



Successful Sim-to-real



Multimodal Predictive Learning
from Interaction in simulation



Reusable Synesthetic Prior
that reasons and predicts the world



Robust Dexterous Control
generalizable, perturbation-robust, sim2real-ready